MILITARY SPECIFICATION

CUSHION COMPONENTS FOR METAL CONTAINER FOR CARTRIDGE 120MM TANK AMMUNITION

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions and packaging for one type of cushion components used for 120MM tank ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicictation.

SPECIFICATIONS

FEDERAL

PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations additions deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U S Army Armament Research Development and Engineering Center Attn AMSMC-QA Dover New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH,DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9390367 - Sleeve, Cartridge Case
9390375 - Base and Nose Cushion (Foam)
9390377 - Support, Projectile, Molded
12631091 - Spacer, Cushion, Foam

(Copies of specifications, standards and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 - Commercial Packaging Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Components. The components shall comply with all requirements specified on drawings 9390375, 9390367, 9390377 or 12631091 and the drawings and specification referenced thereon.

3.3 Water absorption. Water absorption of the polyethylene foam parts, at time of packing, shall not exceed 0.08 pounds per square foot of cut surface when tested in accordance with PPP-C-1752.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. The components shall be regular, smooth and free from any creases, wrinkles, cracks, rips, tears, dirt and any other defect which might offset the servicability, durability, safety and appearance of the assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of ten (10) each of every component.

4.3.2 Inspections to be performed. See MIL-A-48078.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70473 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Base and Nose Cushion (Dwg. 9390375)				
	Examination for Defects	10		3.2	4.4.2.2
	Water Absorption			3.3	4.5.1
	Sleeve, Cartridge Case (Drawing 9390367)				
	Examination for Defects	10		3.2	4.4.2.1
	Water Absorption	10		3.3	4.5.1
	Support Projectile, Molded (Drawing 9390377)				
	Examination for Defects	10		3.2	4.4.2.3
	Spacer, Cushion Foam (Drawing 12631091)				
	Examination for Defects	10		3.2	4.4.2.4
NOTES:					

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of containers shall contain material from lots with the same interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70473 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390367
4.4.2.1	Sleeve, Cartridge Case				NEXT HIGHER ASSEMBLY 9386832 or 9386833
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Material thickness		1.0%	3.2	Gage
103	Outside diameter		1.0%	3.2	Gage
104	Water absorption	4.4.3	1	3.2	4.5.1
105	Density of material	4.4.3	2	3.2	4.5.2
106	Gaps in thermobond		1.0%	3.2	Visual
Minor					
201	Overall length		1.0%	3.2	Gage
202	Poor workmanship		1.0%	3.6	Visual

NOTES:

(a) With each lot of material submitted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70473 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390375
4.4.2.2	Base and Nose Cushion, Foam				NEXT HIGHER ASSEMBLY 9386832 or 9386833
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Thickness		1.0%	3.2	Gage
103	Density of material	4.4.3.2		3.2	Weight Gage
104	Water absorption	4.4.3.1		3.2	4.5.1
Minor					
201	Outside diameter		1.0%	3.2	Gage
202	Number of composite layers		1.0%	3.2	Visual
203	Poor workmanship		1.0%	3.5	Visual

NOTES: (a) With each lot of material submitted.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70473 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Support, Projectile, Molded		9390377
			NEXT HIGHER ASSEMBLY
			9386832 or 9386833

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Overall length		1.0%	3.2	Gage
103	Length of penetrator cup		1.0%	3.2	Gage
104	Diameter of sabot cup		1.0%	3.2	Gage
105	Inside diameter of penetrator cup (spike)		1.0%	3.2	Gage
106	Thickness		1.0%	3.2	Gage
107	Outside diameter, end of penetrator cup (spike)		1.0%	3.2	Gage
108	Width of notch		1.0%	3.2	Gage
109	Distance between notches		1.0%	3.2	Gage
Minor					
201	Angle between fins	(b)	1.0%	3.2	Gage
202	Recess at bottom of sabot cup	(b)	1.0%	3.2	Gage
203	Radius at entrance to sabot cup	(b)	1.0%	3.2	Gage
204	No sprue or flash		1.0%	3.2	Visual
205	Fin Thickness		1.0%	3.2	Gage
206	Location of slots	(b)	1.0%	3.2	Gage
207	Location of notches	(b)		3.2	Gage
208	Length of slots		1.0%	3.2	Gage
209	Width of slots		1.0%	3.2	Gage
210	Outside diameter of slots		1.0%	3.2	Gage
211	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:
(a) With each lot of material submitted.
(b) Three pieces per cavity mold check and after a mold re-work affecting applicable dimensions.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70473 (AR)

PARAGRAPH	TITLE		SHEET 1 of 1		DRAWING NUMBER
4.4.2.4	Spacer, Cushion, Foam				12631091
					NEXT HIGHER ASSEMBLY
					12630717 or 12630718
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Density	(a)		3.2	Certification
Minor					
201	Thickness		1.0%	3.2	Gage
202	Outside diameter		1.0%	3.2	Gage
203	Poor workmanship		1.0%	3.5	Visual

NOTES: (a) With each lot of material submitted

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70473 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Strap and Patch Assembly				12561218
					NEXT HIGHER ASSEMBLY 12561228
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Location of buckle on end of strap		1.0%	3.2	Gage
103	Location of patch from buckle		1.0%	3.2	Gage
104	Location of three patches		1.0%	3.2	Gage
Minor					
201	Stitching not complete		1.0%	3.2	Visual
202	Poor workmanship		1.0%	3.5	Visual
NOTES:	(a) With each lot of material submitted				

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4.4.3 Testing.

4.4.3.1 Water absorption. (See Dwg. 9390375 and 9390367), Major Defect. Three samples per material lot shall be selected prior to start of production. If any sample fails to meet the requirements, the lot shall be rejected.

4.4.3.2 Density. (See Dwg. 9390375 and 9390367), Major Defect. Three samples per material lot shall be selected prior to start of production. If any sample fails to meet the requiremnets, the lot shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.5 herein.

4.5 Methods of inspection.

4.5.1 Water absorption. The water absorption shall be determined in accordance with PPP-C-1752.

4.5.2 Density The density shall be determined in accordance with PPP-C-1752.

5. PACKAGING

5.1 Preservation and packaging. None required.

5.2 Packing.

5.2.1 Level A. The cushion components shall be packed in accordance with Level A of MIL-STD-1169.

5.2.2 Level B. The cushion components shall be packed in accordance with Level B of MIL-STD-1169.

5.2.3 Commercial. The cushion components shall be packed in accordance with ASTM-D-3951.

5.3 Marking.

5.3.1 Level A. The cushion components shall be marked in accordance with Level A of MIL-STD-1169.

5.3.2 Level B. The cushion components shall be marked in accordance with Level B of MIL-STD-1169.

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5.3.3 Commercial. The cushion components shall be marked in accordance with ASTM-D-3951 and the following additional instructions. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, lot number and date, name and address of manufacturer. If components and assemblies are bundled, boxed or palletized, the gross weight and quantity shall be included. The above information may be placed on pallets or boxes by labeling or stenciling.

5.4 Shipping. When cushion components from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

6. NOTES

6.1 Intended use. These cushion components are intended for use as the interior packaging and cushioning for the 120MM Tank Cartridges.

6.2 Ordering data. See MIL-A-48078.

6.3 Definition of packaging and packing levels.

6.3.1 Level A. Full Military protection, for unrestricted (OCONUS) shipment and storage

6.3.2 Level B. Limited Military protection for limited storage and (OCONUS) shipment.

6.3.3 Level C. For interplant (CONUS) shipment and very limited storage.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Submission of inspection equipment for design approvals - See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I(D), Dover, New Jersey 07801-5001.

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6.6 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I(D), Dover, New Jersey 07801-5001. This description should include but not be limited to the accuracy and precision and drawings of any special equipment required.

6.7 Subject term (key word) listing.

Ammunition
Cannon
Cartridge
Container
Cushion Componets
Metal
Tank

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A638)

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70473	CUSHION COMPONENTS FOR METAL CONTAINER FOR CARTRIDGE 120MM TANK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
AMMUNITION	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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